

Follow up

Predicting First-Year Statistics Grades from Attendance, Honesty, and Quiz Results Cohort 2025/29

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Introduction

As of this February, i started as a Teaching Demonstrator at UGD Shtip. Because of wanting to make the exercise sessions more interactive and to gauge pre-course knowledge on Statistics, unintentionally this experiment was started.

Also in order to have an objective way (numeric) of grading the attendance on exercise sessions, homework i started the dataset used in this experiment.

The grade structure at UGD Shtip is as follows:

My points for the subject Financial markets and institutions

Финансиски пазари и институции						
Предавања	Вежби	Проектна задача	Колоквиум 1	Колоквиум 2	Дополнителна активност	Вкупно поени
10	10	10	19	20		69,0

As a Teaching Demonstrator for the cohort i am grading the Exercise Session Points, Course project & Additional Activity.

What Data is collected? 1/5

Attendance is based on three attendance measures:

- Official Attendance list - Faculty demanded
- Paper Attendance list - Rewritten every lecture
- Digital Quiz Attendance Gathering.



20% of the 10 points for Exercise Session is achieved through these three compasses - an Average with cutoffs for each mid-term.

What Data is collected? 2/5

Homework related data: Almost after each lecture with the professor, homework is given which is the other 80% of the total 10 points for Exercise session:

- Number of homework written
- Hand written homework dummy
- Late submission of homework dummy

What Data is collected? 3/5

Every exercise session, mid session there is a google forms quiz from which apart from the mentioned attendance the following data is extracted:

- Quiz score - measure for student mastery of the material.
- Score of hardness of homework written
- Honesty about homework written binary variable - comparing answers with factuals.

What Data is collected? 4/5

Other data derived from the previously mentioned one:

- Sitting close to .25 percentile - low test score and .75 percentile - high test score influence.
- Honesty about the attendance - altering attendance records on the official attendance list.
- Suspected cheating - during the mid-term exam students who turned around or tried forms of workaround, but not doing a misdemeanor to result in a test being taken and having a 3 exam session detention of taking the exam. (approximately 1 year)

What Data is collected? 5/5

As mentioned on the first introductory slide, a background and statistical quiz was performed in order to measure the general knowledge of the audience and to adapt the material regarding it. From which the following background data was taken:

- High school graduation type - economic / law specialized, gymnasium, other types of specialized High school
- Major enrolled - Five types of majors
- Attitude towards Math - 1 - 6 scale
- Passed Math - binary

How is the data collected? 1/3

Name and number of Index (example.084000) *

Short answer text
.....

Did I write homework? *

1. Yes
2. No

How hard was the homework? *

	1	2	3	4	5	6	
Impossible	☐	☐	☐	☐	☐	☐	To easy

How is the data collected? 2/3

Number of index - format (same number of index) example: pete 55555 *

Short answer text
.....

I graduated *

1. Gymnasium
2. Specialized High School - social sciences - (economic / law)
3. Specialized High School - Other

My Major is: *

1. Finance
2. Business Economics
3. Management and Marketing
4. Accounting and Audit
5. International economics and business
6. Other

I consider mathematics to be: *

	1	2	3	4	5	6	
Awful	☐	☐	☐	☐	☐	☐	Favorite

I passed Mathematics for Economists with an: *

	0	1	2	3	4	5	
F	☐	☐	☐	☐	☐	☐	A

How is the data collected? 3/3

Anon index	02.23index	Sits to below .25	Sits to above .75	03.02index	Sits to below .21	Sits to above .71	03.09index	Sits to below .21	Sits to above .71	03.16index	Sits to below .21	Sits to above .71	Mid term	Flag	Final.25	Final .75
21	1	0	1	10	0	1	NA	NA	NA	9	1	0	0.625		0.33	0.67
19	2	0	1	11	1	1	NA	NA	NA	10	1	1	0.525		0.67	1.00
47	3	0	1	40	1	1	NA	NA	NA	NA	NA	NA	0.775		0.50	1.00
48	4	0	1	39	1	1	NA	NA	NA	NA	NA	NA	0.675		0.50	1.00
49	5	0	1	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.35		0.00	1.00
33	6	0	1	28	0	1	18	1	0	NA	NA	NA	0.525		0.33	0.67
32	7	0	1	33	0	0	20	1	1	NA	NA	NA	0.35		0.33	0.67
41	8	0	0	31	0	1	17	1	0	NA	NA	NA	0.425		0.33	0.33
43	9	0	0	34	0	0	NA	NA	NA	NA	NA	NA	NA		0.00	0.00
22	10	1	1	8	0	1	10	0	1	NA	NA	NA	0.575		0.33	1.00
28	11	1	1	9	0	1	8	0	1	NA	NA	NA	0.325		0.33	1.00
6	12	1	1	37	1	1	NA	NA	NA	28	1	1	0.55		1.00	1.00
13	13	1	0	41	1	1	21	1	1	24	1	1	0.125		1.00	0.75
14	14	1	0	42	1	0	NA	NA	NA	26	1	0	0.25		1.00	0.00
54	15	1	0	38	1	1	NA	NA	NA	25	1	0	0.275		1.00	0.33
35	16	1	0	25	1	0	30	1	0	NA	NA	NA	0.4		1.00	0.00
38	17	1	0	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.3		1.00	0.00
42	19	1	0	NA	NA	NA	15	1	0	30	1	1	0.4		1.00	0.33
18	20	1	0	27	0	1	11	1	1	NA	NA	NA	0.4		0.67	0.67
39	21	1	0	24	1	0	24	1	1	11	0	1	0.275		0.75	0.50
4	22	1	0	NA	NA	NA	25	1	1	15	1	1	0.45		1.00	0.67
12	23	1	0	18	1	1	16	1	0	23	1	1	0.25		1.00	0.50
44	24	1	0	26	1	0	26	1	0	27	1	1	0.3		1.00	0.25
3	25	0	0	23	1	0	29	1	0	NA	NA	NA	0.525		0.67	0.00
50	26	0	1	17	1	1	NA	NA	NA	20	0	1	NA		0.33	1.00
34	27	1	1	16	1	0	NA	NA	NA	21	1	1	NA		1.00	0.67
30	28	1	1	20	1	1	9	0	1	22	1	0	0.575		0.75	0.75
46	29	1	1	21	0	1	33	1	1	14	0	1	0.475		0.50	1.00
26	30	1	1	19	1	1	34	0	0	13	1	0	0.6		0.75	0.50
16	31	1	1	14	1	0	31	1	0	17	0	0	0.275		0.75	0.25
5	32	1	1	13	1	0	32	1	1	16	1	0	0.525		1.00	0.50
7	33	1	0	NA	NA	NA	7	0	1	NA	NA	NA	NA		0.50	0.50
53	34	1	0	4	0	1	NA	NA	NA	6	0	1	0.325		0.33	0.67
25	35	0	0	2	0	0	2	1	0	2	1	1	0.525		0.50	0.25
23	36	0	0	3	0	1	3	1	1	3	1	1	0.25		0.50	0.75
15	37	0	1	1	0	0	4	0	1	9	0	1	0.525		0.00	0.75
38	1	1	7	0	1	6	0	1	7	0	1	0.55		0.25	1.00	
52	39	1	1	6	0	1	5	0	0	4	0	0	0.775		0.25	0.50
55	40	0	0	5	0	0	NA	NA	NA	NA	NA	NA	NA		0.00	0.00

Hypotheses

- H0 Attendance, homework completion/timeliness, and academic honesty predict test performance more strongly than attitude toward math.
- H1 Seating proximity to lower- or higher-achieving peers affects student performance compared with students near the median.
- H2 Lower attendance, late homework submission, digitally completed homework, and low quiz scores are associated with lower final results, and the opposite patterns predict higher results.

Statistical Methods

- Regressions
- ART-2a clustering

Hypothesis 0

- H0 Attendance, homework completion/timeliness, and academic honesty predict test performance more strongly than attitude toward math.

$$Y = \text{averATND} + \text{numHW} + \text{paperHW} + \text{lateHW} \\ + \text{additHW} + \text{honLST} + \text{honHW} + \text{atitMATH}$$

Hypothesis 0

- Using OLS with HW errors statistically significant were only the group average Attendance and if the person was suspected of Cheating.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.159647	0.183254	6.328075	0.0000
group	-0.151776	0.045661	-3.323938	0.0038
averATND	-0.140418	0.049650	-2.828162	0.0111
numHW	-0.021964	0.064251	-0.341845	0.7364
paperHW	0.143849	0.103936	1.384014	0.1833
lateHW	0.199300	0.132025	1.509566	0.1485
CHEAT	-0.181898	0.052942	-3.435793	0.0029
atitMATH	0.018119	0.014738	1.229392	0.2348
R-squared	0.624018	Mean dependent var	0.463462	
Adjusted R-squared	0.477802	S.D. dependent var	0.163130	
S.E. of regression	0.117883	Akaike info criterion	-1.190583	
Sum squared resid	0.250137	Schwarz criterion	-0.803476	
Log likelihood	23.47758	Hannan-Quinn criter.	-1.079110	
F-statistic	4.267800	Durbin-Watson stat	2.031867	
Prob(F-statistic)	0.006096	Wald F-statistic	10.47473	
Prob(Wald F-statistic)	0.000031			

Hypothesis 0a

- The modified OLS model with HW errors statistically significant we still can conclude that at the cut of of 5% attitude towards Mathematics is statistically insignificant.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.096379	0.171329	6.399253	0.0000
group	-0.173178	0.041489	-4.174036	0.0004
averATND	-0.105257	0.045496	-2.313561	0.0304
CHEAT	-0.156560	0.054328	-2.881771	0.0087
atitMATH	0.025789	0.013545	1.903924	0.0701
R-squared	0.564489	Mean dependent var	0.455556	
Adjusted R-squared	0.485305	S.D. dependent var	0.165153	
S.E. of regression	0.118485	Akaike info criterion	-1.262489	
Sum squared resid	0.308850	Schwarz criterion	-1.022519	
Log likelihood	22.04360	Hannan-Quinn criter.	-1.191133	
F-statistic	7.128835	Durbin-Watson stat	1.793407	
Prob(F-statistic)	0.000771	Wald F-statistic	10.33381	
Prob(Wald F-statistic)	0.000073			

Hypothesis 0c

- From the clustering using ART-2a comparing medians we could notice that except from the outlier in cluster 3, we can conclude that in general across the clusters other factors are more deterministic of the test scores than the reported attitude towards Mathematics.

H0	ρ^*	0.8	β	0.85					
Cluster	num. of obs	Y	MATH	ATND	HW	QIZZ	pHW	hLST	hHW
MEDIAN	NA	0.45	3	3	3	0.33	0.67	1	1
0	2	0.2625	1	<u>3.33</u>	0.5	0.2949	0.1667	1.0	1
1	3	0.425	2	<u>4</u>	3	<u>0.6431</u>	<u>1</u>	1.0	1.0
2	4	0.4	2.5	<u>3.33</u>	1	0.2541	0.1667	1.0	1
3	1	<u>0.55</u>	<u>5</u>	2	1	0.28	0.3333	1.0	1.0
4	17	<u>0.525</u>	3	3	3	<u>0.3521</u>	<u>1</u>	1.0	1.0

Hypothesis 1

- H1 Seating proximity to lower- or higher-achieving peers affects student performance compared with students near the median.

$$Y = c + \textit{proxi}.25 + \textit{proxi}.75$$

Hypothesis 1

- Sitting close to the lower 25% using the OLS model results in a statistically insignificant results, on the other hand sitting to an student which has a result in the upper 75% results in an increase of an average increase of 16%.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.566482	0.104215	5.435677	0.0000
GROUP	-0.109507	0.042365	-2.584854	0.0133
PROXI_25	-0.094443	0.074474	-1.268141	0.2117
PROXI_75	0.163784	0.066376	2.467520	0.0178
R-squared	0.321135	Mean dependent var	0.440761	
Adjusted R-squared	0.272645	S.D. dependent var	0.163415	
S.E. of regression	0.139369	Akaike info criterion	-1.020449	
Sum squared resid	0.815791	Schwarz criterion	-0.861436	
Log likelihood	27.47032	Hannan-Quinn criter.	-0.960882	
F-statistic	6.622660	Durbin-Watson stat	1.725964	
Prob(F-statistic)	0.000915	Wald F-statistic	7.685862	
Prob(Wald F-statistic)	0.000333			

Hypothesis 1a

- Adding the logic dummy of suspected cheating into the OLS model, results in an insignificant effect of it.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.580085	0.122129	4.749762	0.0000
GROUP	-0.109421	0.042601	-2.568509	0.0140
PROXI_25	-0.093502	0.074389	-1.256925	0.2159
PROXI_75	0.164181	0.067883	2.418584	0.0201
CHEAT	-0.019142	0.050753	-0.377164	0.7080
R-squared	0.323684	Mean dependent var	0.440761	
Adjusted R-squared	0.257702	S.D. dependent var	0.163415	
S.E. of regression	0.140793	Akaike info criterion	-0.980732	
Sum squared resid	0.812728	Schwarz criterion	-0.781967	
Log likelihood	27.55685	Hannan-Quinn criter.	-0.906274	
F-statistic	4.905643	Durbin-Watson stat	1.755268	
Prob(F-statistic)	0.002517	Wald F-statistic	6.565955	
Prob(Wald F-statistic)	0.000352			

Hypothesis 1c-nc

- From the students not suspected of cheating we have the following clusters:

H1-noncheater	ρ^*	0.65	β	1
Cluster	number of obs	Y	Proxi.25	Proxi.75
MEDIAN	NA	0.45	0.42	0.88
0	3	0.15	0	0
1	4	0.35	1	0
2	4	<u>0.5125</u>	0.415	0.875
3	2	<u>0.5625</u>	0.875	0.875

Hypothesis 1c-ch

- From the students suspected of cheating we have the following clusters:

H1-cheater	ρ^*	0.65	β	1
Cluster	number of obs	Y	Proxi.25	Proxi.75
MEDIAN	NA	0.47	0.67	0.5
0	14	<u>0.3125</u>	1	0.33
1	4	<u>0.375</u>	1	0.875
2	12	<u>0.5</u>	0.33	<u>1</u>
3	4	0.4875	0.415	0.415
4	2	<u>0.525</u>	0.585	0.125

Hypothesis 2

- H2 Lower attendance, late homework submission, digitally completed homework, and low quiz scores are associated with lower final results, and the opposite patterns predict higher results.

$$Y = \text{averATND} + \text{averQIZZ} + \text{numHW} + \\ + \text{paperHW} + \text{lateHW} + \text{hardHW}$$

Hypothesis 2

- Using an OLS model, non of the purposed variables have an statistically significant effect on the results on the Midterm exam.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.702047	0.133771	5.248139	0.0000
GROUP	-0.110862	0.067779	-1.635643	0.1120
AVERATND	-0.071208	0.040783	-1.746003	0.0907
AVERQIZZ	0.164981	0.257482	0.640750	0.5264
NUMHW	0.016727	0.053354	0.313518	0.7560
PAPERHW	0.027505	0.115557	0.238021	0.8134
LATEHW	-0.050928	0.132789	-0.383522	0.7040
HARDHW	0.047602	0.202960	0.234540	0.8161
R-squared	0.206920	Mean dependent var	0.452564	
Adjusted R-squared	0.027837	S.D. dependent var	0.165513	
S.E. of regression	0.163193	Akaike info criterion	-0.607083	
Sum squared resid	0.825591	Schwarz criterion	-0.265840	
Log likelihood	19.83813	Hannan-Quinn criter.	-0.484648	
F-statistic	1.155443	Durbin-Watson stat	0.975067	
Prob(F-statistic)	0.355879	Wald F-statistic	1.387053	
Prob(Wald F-statistic)	0.245763			

Hypothesis 2c

- The clustering results show us the following clusters:

H2	ρ^*	0.65	β	1				
Cluster	number of obs	Y	averATND	averQIZZ	numHW	paperHW	lateHW	hardHW
MEDIAN	NA	0.4	3	0.3	2	0.33	0	0.39
0	3	0.15	0	0	0	0	0	0
1	36	<u>0.4375</u>	3	<u>0.3194</u>	3	<u>0.6667</u>	0	<u>0.5</u>
2	13	0.275	3	0.2204	1	0	0	0.2778
3	2	<u>0.5125</u>	<u>0.6667</u>	0.1	2	<u>0.5</u>	0	<u>0.0833</u>

Limitations

• Exam bias

The initial experiment design was to have a separate person (the professor to examine the mid-term exams) to mitigate for potential bias. But unfortunately this was not applied as the author and the examiner of the exams was me, but as there is little to none space for bias induced error.

This is not a big limitation because of the nature of the subject being solution to exercises, not a matter of explanation of a theory and interpretation of natural language which could induce bias by the examiner. (such as unreadable handwriting, wording, etc.)

• Small sample size

In fact the sample size is the whole population, the statistical methods mostly have remedies to handle these drawbacks of small sample case of 49 students who took the midterm. Such as reshuffling to get consistent clusters in the case of ART-2a

Ending remarks

I am delighted for your attention.

Q & A?